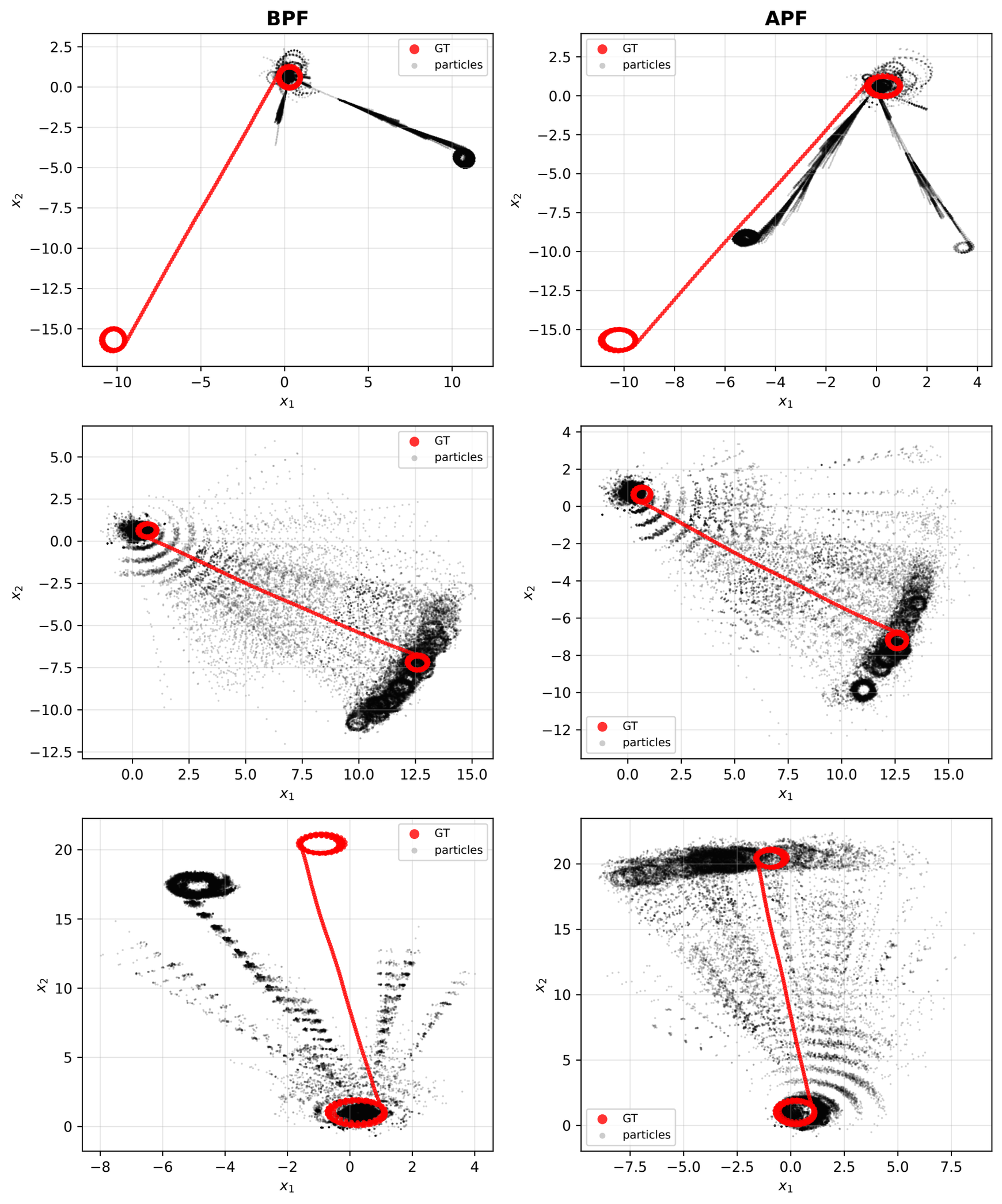

BPF
APF
GT
particles
x_1
x_2